MIL-H-37011A
1 December 1976
SUPERSEDING
MIL-H-37011
2 October 1973

MILITARY SPECIFICATION

HOLDER, SUTURE NEEDLE, DeBAKEY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of suture needle holder of the DeBakey design for use in open heart surgery.

1.2 Classification.

1.2.1 Sizes. The holder shall be of the following sizes as specified (see 6.2):

Size		Length (in inches)
1	-	7
2	-	9
3	-	10
4	-	12

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526 - Instruments, Dental and Surgical; General Specifications.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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SPECIFICATIONS

Federal

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Set-up.

Military

MIL-P-116	-	Preservation, Methods of.
MIL-L-10547	-	Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of specifications and standards, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

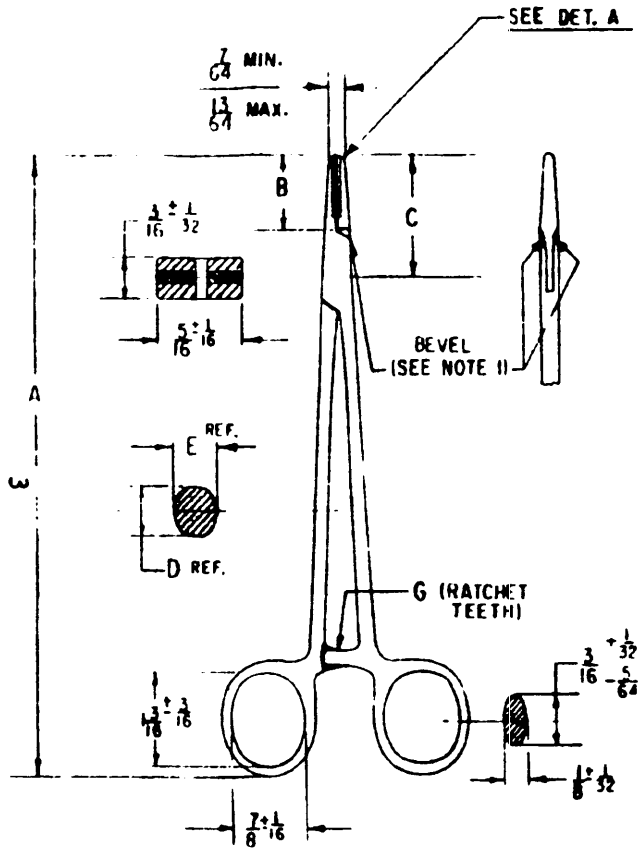
3.1 General requirements. The holder shall be in accordance with the applicable requirements of GG-I-526 and the following specific requirements:

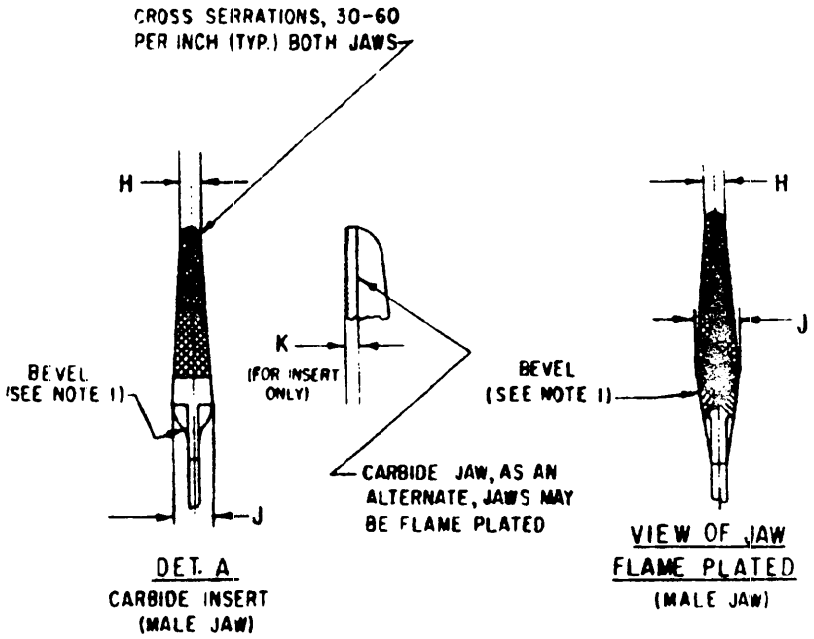
3.2 Material. The holder proper shall be fabricated from Class 4, Type 420 material. Jaw inserts (or material plated onto the inside of the jaws) shall be tungsten carbide.

3.3 Hardness. After proper stress relieving, the holder shall have a Rockwell hardness of C40 to C45, with the exception of the screw or pin, for the holder proper. Tungsten carbide insert shall have a Rockwell hardness of A80 minimum when removed from the jaw of a completed instrument.

3.4 Style, design and dimensions. Shall be a cardiac needle holder of the DeBakey design. Serrated inserts shall be brazed to the jaw with a uniform flow of silver solder and be free of crevices at all interfaces. As an alternate, the serrated inside surfaces of the jaws shall be flame plated with a deposit of tungsten carbide material. Thickness of plating shall be 0.004 ± .001 inch. Shall be in accordance with Figure 1.

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SIZE	NSN	A	B ± 1/16	C ± 1/16	D REF.	E REF.	G	H ±.012 BOTH JAWS	J BOTH JAWS	K ±.003
1	6515-00-690-3198	7 1/16 ± 3/8	7/8	1 1/4	7/64	1/8	3x3 or 4x4	.069	.156 ±.041	.015
2	6515-00-690-3200	9 ± 1/4	1	1 5/16	7/64	1/8	4x4	.069	.166 ±.032	.015
3	6515-00-690-3201	10 5/16 ± 1/2	1	1 7/16	11/64	5/32	4x4	.069	.176 ±.022	.031
4	6515-00-690-3202	12 ± 1/4	1	1 7/16	11/64	5/32	4x4	.069	186 ±.012	.031

NOTES:

1. BEVEL SHALL BE PROVIDED ON BOTH SIDES OF THE DISTAL EDGE OF THE MALE MEMBER OF THE JAW, ADJACENT TO THE LOCK.
2. TOLERANCES INDICATED DELINEATE ACCEPTABLE EXTREME DIMENSIONS AND ARE NOT INTENDED TO PERMIT VARIATIONS WITHIN A SINGLE INSTRUMENT WHICH WILL SACRIFICE UNIFORMITY, SYMMETRY, AND FIRST QUALITY WORKMANSHIP.

FIGURE 1. Holder, suture needle.

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3.4.1 Locks. Shall be box type lock.

3.5 Finish. Finish shall be mirror or satin for the holder except the inserts. Locks and ratchets shall be smooth and shall be polished, but need not be buffed to a mirror finish.

3.6 Lock stiffness. Holder shall comply with requirements when tested in accordance with GG-I-526.

3.7 Identification marking. In addition to GG-I-526 requirements, holders of foreign manufacture shall also be marked "stainless".

3.8 Needle retention. Holder shall be suitable for securely holding suture needles, when tested as specified in 4.4.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection shall be in accordance with GG-I-526.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the corrosion resisting steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one holder.

TABLE I. Sampling criteria for examination.

Type of defect	Inspection level	AQL (Percent defective)
Major	II	1.0
Minor	II	2.5

* 4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
1-Holder	Hardness	3.3 and GG-I-526	GG-I-526	S-2	1.5
	Dimensions including clearances (crevices)	3.4	------	S-2	2.5
	Boil test	GG-I-526	GG-I-526	S-2	1.0
	Copper sulfate	GG-I-526	GG-I-526	S-2	1.0
	Box lock stiffness	3.6	GG-I-526	S-2	2.5
	Needle retention	3.8	4.4.1	S-2	1.0

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the manufacture of the holders shall be selected for composition analysis in accordance with GG-I-526.

* 4.3 Examination. The holder shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any holder in the sample containing one or more defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

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TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Holder not of box lock construction.
102	Holder not free from excessive play in box lock area.
103	Ratchet and ratchet catches do not easily engage and disengage at each ratchet position.
104	Ratchet and ratchet catches do not lock securely.
Minor	
201	Holder not identified as specified.
202	Holder not free from foreign matter (polishing material).
203	Finish not as specified.

4.3.1 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

* 4.4.1 Needle retention test. The longitudinal axis of a 25 ml (0.025 inch) diameter suture needle shall be held perpendicular to the axis of the jaws, anywhere along the length of the jaw serrations and the third position of the ratchet engaged. Alternately, the suture needle specified may be placed on a plane surface so that its longitudinal axis lies along the axis of the plane. With the ratchet closure disengaged the instrument shall be positioned so that the jaw tips will grasp the needle upon engagements of the ratchets in either method. With the ratchet engaged in the third position and the jaws clamped on the needle, there shall be no visible lateral rotational or longitudinal movement of the needle nor shall there be any lateral displacement of the jaws. Upon disengagement of the ratchets and removal of the needle, there shall be no visible distortion of the instrument jaws.

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each holder shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.1.2 Preservative. Preservative shall not be applied.

5.1.1.3 Unit packaging. Each holder shall be wrapped in neutral tissue or other suitable anticorrosive materials. Pliers shall be further packaged in accordanc with MIL-P-116, method IC-1.

5.1.4 Intermediate packaging. Twelve (12) holders shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566, PPP-B-636, type CF, class domestic or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. One hundred forty four (144) holders shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, type CF, class weather resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. One hundred forty four (144) holders shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636 type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. Shall be a needle holder of the DeBakey design which is specifically designed for open heart surgery.

6.2 Ordering data. Procurement documents should specify the following.

(a) Number, title and date of this specification.
(b) Size required (see 1.2.1).
(c) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification covers the following items listed in the Federal Supply Catalog:

National Stock No.	Item Identification	Size
6515-00-690-3198	HOLDER, SUTURE NEEDLE, DeBakey, 7 inch	1
6515-00-690-3200	HOLDER, SUTURE NEEDLE, DeBakey, 9 inch	2
6515-00-690-3201	HOLDER, SUTURE NEEDLE, DeBakey, 10 inch	3
6515-00-690-3202	HOLDER, SUTURE NEEDLE, DeBakey, 12 inch	4

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

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6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - MD
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Preparing activity:
Defense Logistic Agency-DM

Project No. 6515-1983

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-H-37011A HOLDER, SUTURE NEEDLE, DeBAKEY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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